

8DM60: Multimodal AI in Biomedicine

Lecture 2: Guided self-study (BZ)

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Problem 1

A biomedical AI researcher trains a DINO-style student and teacher on unlabelled microscopy images. After each student gradient update, the teacher is updated by

$$\theta_{\text{teacher}} \leftarrow m\theta_{\text{teacher}} + (1 - m)\theta_{\text{student}}.$$

Compare four fixed choices: $m = 0$, $m = 0.9$, $m = 0.999$, and $m = 1$. Explain how each affects the teacher's response to changes in the student. Why might using a slowly changing teacher be useful?

Problem 2

A pathology researcher removes feature-spread regularization from DINOv2 pretraining. The image-level and patch-level matching objectives, teacher update, and target balancing remain unchanged.

Here, matching loss is the cross-entropy between the teacher's target distribution and the student's predicted distribution, applied to image-level outputs and corresponding patch-level outputs.

How could this affect the learned representations? Describe what you would compare between training runs with and without the regularizer to determine whether the features remain useful. Is matching loss alone sufficient?

Problem 3

Two encoders are evaluated on the same held-out tissue images. Both use the same training labels. Within each evaluation method, the head architecture and training budget are comparable.

Encoder	Frozen encoder + linear head: accuracy	Fine-tuned encoder + head: accuracy
A	78%	86%
B	84%	85%

Which encoder does the table favour when only a linear head can be trained on saved features? Which does it favour when the encoder can also be updated? Explain why these rankings can differ and what the table alone cannot establish.

Problem 4

A teaching-image classifier applies softmax to CLIP similarities. With the candidate descriptions “normal” and “abnormal,” the score for “abnormal” is 0.90. An engineer adds a third description without changing the model.

Could the score for “abnormal” change even if its similarity to the image stays the same? Explain what this means for interpreting 0.90 as a patient’s probability of disease.

Problem 5

A CLIP-style training batch contains two chest images with identical paired text: “No visible abnormality.” The deterministic text encoder produces the same vector for both copies. Each image is trained to select its own recorded text over every other text in the batch.

Can either image assign all probability to its recorded copy? Explain the conflict and propose one change to the objective or batch construction.

Problem 6

A radiology educator wants to search an image archive using text and generate new explanations of selected images. He uses a frozen CLIP model for search. For explanations, he passes its image vector directly to a frozen text-only language model as one input embedding, followed by the user's question. The vector has the required length; the two models have not been trained together.

Assess both parts of this design. Is matching the vector length enough for the explanation function? Describe any training needed to connect the models.

Problem 7

A laboratory needs one output per microscopy image: “usable” or “unusable.” It has representative labels and a fixed inference budget. Option A fits a small head on a frozen visual encoder. Option B instruction-tunes a visual language model to produce the same two labels. Both meet the current accuracy target.

Compare their advantages and disadvantages for this task. What additional measurements would you use to choose between them?

Problem B1 (bonus)

A hospital uses a frozen CLIP-style model to sort teaching images into “X-ray,” “CT,” and “MRI” by comparing images with text descriptions. It now wants to include “ultrasound.”

What must change to include the new category? Is retraining required to produce a prediction, and what should be checked before trusting the new predictions?

Problem B2 (bonus)

After image-caption alignment, a biomedical visual assistant gives broad descriptions even when asked a specific question. Its developers can either add more image-caption pairs or collect specific questions with appropriate answers. Both options cost the same.

Which option more directly addresses this behaviour? Explain what it adds and one limitation it does not automatically solve.

Problem B3 (bonus)

A hospital's visual assistant receives a new chest image, a request to describe it, and captions retrieved from similar archive images. The developers expect these captions to help when the new image is difficult to interpret. They can also run the same assistant without retrieval. Radiologists can review a small held-out image set.

Design a comparison to decide whether to keep retrieval. Specify the test conditions, what radiologists should assess, and how you would distinguish useful context from errors introduced by the retrieved captions.